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EXPERIEMNT 25

Prolog Program: Monkey-Banana Problem

Aim

To write a **Prolog program to implement the Monkey-Banana problem**, where a monkey tries to get a banana hanging from the ceiling by using a box.

Algorithm

1. Represent the positions of the monkey, box, and banana.
2. The monkey can **walk** from one position to another.
3. The monkey can **push the box** to another position.
4. The monkey can **climb onto the box**.
5. When the monkey is on the box and at the banana position, it can **grasp the banana**.
6. Display the sequence of actions required to get the banana.

Prolog Code

```
% Monkey Banana Problem


% Initial state

initial_state(state(door, window, floor, no)).


% Monkey walks from door to window

move(state(door, window, floor, no),
```

```
    state(window, window, floor, no),
    walk).

% Monkey pushes box from window to middle
move(state(window, window, floor, no),
    state(middle, middle, floor, no),
    push).

% Monkey climbs onto the box
move(state(middle, middle, floor, no),
    state(middle, middle, onbox, no),
    climb).

% Monkey gets the banana
move(state(middle, middle, onbox, no),
    state(middle, middle, onbox, yes),
    get_banana).

% Solve the problem
solve(Actions) :-
   initial_state(S0),
   move(S0, S1, A1),
   move(S1, S2, A2),
   move(S2, S3, A3),
```

```
    move(S3, S4, A4),

    S4 = state(middle, middle, onbox, yes),

    Actions = [A1, A2, A3, A4].
```

Input

solve(X).

Output

```
| ?- consult('C:/Users/MOSHIYA/Downloads/Exppp 25.pl').
compiling C:/Users/MOSHIYA/Downloads/Exppp 25.pl for byte code...
C:/Users/MOSHIYA/Downloads/Exppp 25.pl compiled, 33 lines read - 3050 bytes written, 16 ms

yes
| ?- solve(X).

X = [walk,push,climb,get_banana] ?
```

Result

Thus, the Monkey-Banana problem was successfully implemented in Prolog, and the sequence of actions performed by the monkey to obtain the banana was displayed.